

```
In [1]: import numpy as np
import tensorflow as tf
from tensorflow import keras

model = tf.keras.Sequential([keras.layers.Dense(units=1, input_shape=[1])])
model.summary()

model.compile(optimizer="sgd", loss="mean_squared_error")

xs = np.array([-1.0, 0.0, 1.0, 2.0, 3.0, 4.0], dtype=float)
ys = np.array([-2.0, 1.0, 4.0, 7.0, 10.0, 13.0], dtype=float)

model.fit(xs, ys, epochs=500)

print(model.predict(np.array([10.0])))
```

D:\Anaconda\Lib\site-packages\keras\src\layers\core\dense.py:87: UserWarning: Do not pass an `input\_shape`/`input\_dim` argument to a layer. When using Sequential models, prefer using an `Input(shape)` object as the first layer in the model instead.

```
super().__init__(activity_regularizer=activity_regularizer, **kwargs)
```

Model: "sequential"

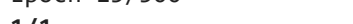
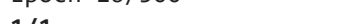
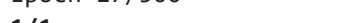
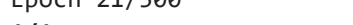
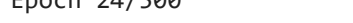
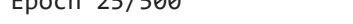
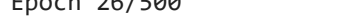
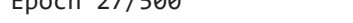
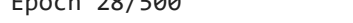
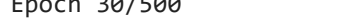
Layer (type)	Output Shape
dense (Dense)	(None, 1)



Total params: 2 (8.00 B)

Trainable params: 2 (8.00 B)

Non-trainable params: 0 (0.00 B)

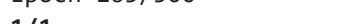
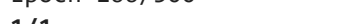
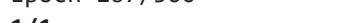
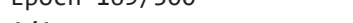
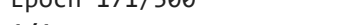
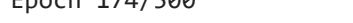
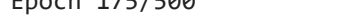
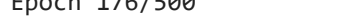
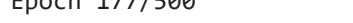
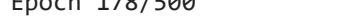
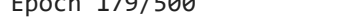
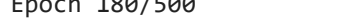
Epoch 1/500
1/1  0s 128ms/step - loss: 44.0941
Epoch 2/500
1/1  0s 13ms/step - loss: 34.6933
Epoch 3/500
1/1  0s 14ms/step - loss: 27.2972
Epoch 4/500
1/1  0s 14ms/step - loss: 21.4783
Epoch 5/500
1/1  0s 11ms/step - loss: 16.9002
Epoch 6/500
1/1  0s 15ms/step - loss: 13.2983
Epoch 7/500
1/1  0s 13ms/step - loss: 10.4645
Epoch 8/500
1/1  0s 14ms/step - loss: 8.2349
Epoch 9/500
1/1  0s 15ms/step - loss: 6.4808
Epoch 10/500
1/1  0s 14ms/step - loss: 5.1006
Epoch 11/500
1/1  0s 15ms/step - loss: 4.0148
Epoch 12/500
1/1  0s 14ms/step - loss: 3.1605
Epoch 13/500
1/1  0s 15ms/step - loss: 2.4883
Epoch 14/500
1/1  0s 14ms/step - loss: 1.9594
Epoch 15/500
1/1  0s 18ms/step - loss: 1.5433
Epoch 16/500
1/1  0s 14ms/step - loss: 1.2158
Epoch 17/500
1/1  0s 13ms/step - loss: 0.9582
Epoch 18/500
1/1  0s 13ms/step - loss: 0.7555
Epoch 19/500
1/1  0s 14ms/step - loss: 0.5959
Epoch 20/500
1/1  0s 13ms/step - loss: 0.4704
Epoch 21/500
1/1  0s 16ms/step - loss: 0.3716
Epoch 22/500
1/1  0s 14ms/step - loss: 0.2938
Epoch 23/500
1/1  0s 13ms/step - loss: 0.2326
Epoch 24/500
1/1  0s 13ms/step - loss: 0.1844
Epoch 25/500
1/1  0s 14ms/step - loss: 0.1465
Epoch 26/500
1/1  0s 15ms/step - loss: 0.1166
Epoch 27/500
1/1  0s 14ms/step - loss: 0.0931
Epoch 28/500
1/1  0s 14ms/step - loss: 0.0745
Epoch 29/500
1/1  0s 14ms/step - loss: 0.0599
Epoch 30/500
1/1  0s 19ms/step - loss: 0.0484

Epoch 31/500
1/1  0s 13ms/step - loss: 0.0393
Epoch 32/500
1/1  0s 14ms/step - loss: 0.0321
Epoch 33/500
1/1  0s 14ms/step - loss: 0.0264
Epoch 34/500
1/1  0s 15ms/step - loss: 0.0219
Epoch 35/500
1/1  0s 14ms/step - loss: 0.0184
Epoch 36/500
1/1  0s 17ms/step - loss: 0.0156
Epoch 37/500
1/1  0s 15ms/step - loss: 0.0133
Epoch 38/500
1/1  0s 14ms/step - loss: 0.0115
Epoch 39/500
1/1  0s 14ms/step - loss: 0.0101
Epoch 40/500
1/1  0s 14ms/step - loss: 0.0090
Epoch 41/500
1/1  0s 14ms/step - loss: 0.0081
Epoch 42/500
1/1  0s 13ms/step - loss: 0.0073
Epoch 43/500
1/1  0s 14ms/step - loss: 0.0067
Epoch 44/500
1/1  0s 14ms/step - loss: 0.0062
Epoch 45/500
1/1  0s 15ms/step - loss: 0.0058
Epoch 46/500
1/1  0s 14ms/step - loss: 0.0055
Epoch 47/500
1/1  0s 15ms/step - loss: 0.0052
Epoch 48/500
1/1  0s 13ms/step - loss: 0.0049
Epoch 49/500
1/1  0s 14ms/step - loss: 0.0047
Epoch 50/500
1/1  0s 13ms/step - loss: 0.0045
Epoch 51/500
1/1  0s 13ms/step - loss: 0.0044
Epoch 52/500
1/1  0s 14ms/step - loss: 0.0042
Epoch 53/500
1/1  0s 14ms/step - loss: 0.0041
Epoch 54/500
1/1  0s 15ms/step - loss: 0.0040
Epoch 55/500
1/1  0s 13ms/step - loss: 0.0039
Epoch 56/500
1/1  0s 13ms/step - loss: 0.0038
Epoch 57/500
1/1  0s 14ms/step - loss: 0.0037
Epoch 58/500
1/1  0s 15ms/step - loss: 0.0036
Epoch 59/500
1/1  0s 14ms/step - loss: 0.0035
Epoch 60/500
1/1  0s 13ms/step - loss: 0.0034

Epoch 61/500
1/1  0s 13ms/step - loss: 0.0034
Epoch 62/500
1/1  0s 13ms/step - loss: 0.0033
Epoch 63/500
1/1  0s 14ms/step - loss: 0.0032
Epoch 64/500
1/1  0s 14ms/step - loss: 0.0031
Epoch 65/500
1/1  0s 14ms/step - loss: 0.0031
Epoch 66/500
1/1  0s 14ms/step - loss: 0.0030
Epoch 67/500
1/1  0s 15ms/step - loss: 0.0030
Epoch 68/500
1/1  0s 13ms/step - loss: 0.0029
Epoch 69/500
1/1  0s 14ms/step - loss: 0.0028
Epoch 70/500
1/1  0s 14ms/step - loss: 0.0028
Epoch 71/500
1/1  0s 13ms/step - loss: 0.0027
Epoch 72/500
1/1  0s 15ms/step - loss: 0.0027
Epoch 73/500
1/1  0s 15ms/step - loss: 0.0026
Epoch 74/500
1/1  0s 14ms/step - loss: 0.0025
Epoch 75/500
1/1  0s 13ms/step - loss: 0.0025
Epoch 76/500
1/1  0s 13ms/step - loss: 0.0024
Epoch 77/500
1/1  0s 14ms/step - loss: 0.0024
Epoch 78/500
1/1  0s 14ms/step - loss: 0.0023
Epoch 79/500
1/1  0s 14ms/step - loss: 0.0023
Epoch 80/500
1/1  0s 14ms/step - loss: 0.0022
Epoch 81/500
1/1  0s 13ms/step - loss: 0.0022
Epoch 82/500
1/1  0s 14ms/step - loss: 0.0022
Epoch 83/500
1/1  0s 14ms/step - loss: 0.0021
Epoch 84/500
1/1  0s 14ms/step - loss: 0.0021
Epoch 85/500
1/1  0s 13ms/step - loss: 0.0020
Epoch 86/500
1/1  0s 14ms/step - loss: 0.0020
Epoch 87/500
1/1  0s 14ms/step - loss: 0.0019
Epoch 88/500
1/1  0s 14ms/step - loss: 0.0019
Epoch 89/500
1/1  0s 15ms/step - loss: 0.0019
Epoch 90/500
1/1  0s 14ms/step - loss: 0.0018

Epoch 91/500
1/1  0s 14ms/step - loss: 0.0018
Epoch 92/500
1/1  0s 15ms/step - loss: 0.0018
Epoch 93/500
1/1  0s 15ms/step - loss: 0.0017
Epoch 94/500
1/1  0s 14ms/step - loss: 0.0017
Epoch 95/500
1/1  0s 15ms/step - loss: 0.0016
Epoch 96/500
1/1  0s 14ms/step - loss: 0.0016
Epoch 97/500
1/1  0s 13ms/step - loss: 0.0016
Epoch 98/500
1/1  0s 15ms/step - loss: 0.0015
Epoch 99/500
1/1  0s 15ms/step - loss: 0.0015
Epoch 100/500
1/1  0s 14ms/step - loss: 0.0015
Epoch 101/500
1/1  0s 14ms/step - loss: 0.0015
Epoch 102/500
1/1  0s 16ms/step - loss: 0.0014
Epoch 103/500
1/1  0s 14ms/step - loss: 0.0014
Epoch 104/500
1/1  0s 14ms/step - loss: 0.0014
Epoch 105/500
1/1  0s 14ms/step - loss: 0.0013
Epoch 106/500
1/1  0s 13ms/step - loss: 0.0013
Epoch 107/500
1/1  0s 14ms/step - loss: 0.0013
Epoch 108/500
1/1  0s 14ms/step - loss: 0.0013
Epoch 109/500
1/1  0s 14ms/step - loss: 0.0012
Epoch 110/500
1/1  0s 15ms/step - loss: 0.0012
Epoch 111/500
1/1  0s 15ms/step - loss: 0.0012
Epoch 112/500
1/1  0s 14ms/step - loss: 0.0012
Epoch 113/500
1/1  0s 14ms/step - loss: 0.0011
Epoch 114/500
1/1  0s 14ms/step - loss: 0.0011
Epoch 115/500
1/1  0s 14ms/step - loss: 0.0011
Epoch 116/500
1/1  0s 13ms/step - loss: 0.0011
Epoch 117/500
1/1  0s 15ms/step - loss: 0.0010
Epoch 118/500
1/1  0s 13ms/step - loss: 0.0010
Epoch 119/500
1/1  0s 14ms/step - loss: 0.0010
Epoch 120/500
1/1  0s 14ms/step - loss: 9.8019e-04

Epoch 121/500
1/1  0s 14ms/step - loss: 9.6005e-04
Epoch 122/500
1/1  0s 17ms/step - loss: 9.4034e-04
Epoch 123/500
1/1  0s 15ms/step - loss: 9.2102e-04
Epoch 124/500
1/1  0s 14ms/step - loss: 9.0210e-04
Epoch 125/500
1/1  0s 16ms/step - loss: 8.8357e-04
Epoch 126/500
1/1  0s 14ms/step - loss: 8.6542e-04
Epoch 127/500
1/1  0s 14ms/step - loss: 8.4764e-04
Epoch 128/500
1/1  0s 14ms/step - loss: 8.3023e-04
Epoch 129/500
1/1  0s 14ms/step - loss: 8.1318e-04
Epoch 130/500
1/1  0s 15ms/step - loss: 7.9648e-04
Epoch 131/500
1/1  0s 14ms/step - loss: 7.8012e-04
Epoch 132/500
1/1  0s 14ms/step - loss: 7.6409e-04
Epoch 133/500
1/1  0s 15ms/step - loss: 7.4840e-04
Epoch 134/500
1/1  0s 14ms/step - loss: 7.3302e-04
Epoch 135/500
1/1  0s 14ms/step - loss: 7.1797e-04
Epoch 136/500
1/1  0s 14ms/step - loss: 7.0322e-04
Epoch 137/500
1/1  0s 14ms/step - loss: 6.8877e-04
Epoch 138/500
1/1  0s 13ms/step - loss: 6.7462e-04
Epoch 139/500
1/1  0s 15ms/step - loss: 6.6077e-04
Epoch 140/500
1/1  0s 15ms/step - loss: 6.4720e-04
Epoch 141/500
1/1  0s 14ms/step - loss: 6.3390e-04
Epoch 142/500
1/1  0s 13ms/step - loss: 6.2088e-04
Epoch 143/500
1/1  0s 18ms/step - loss: 6.0812e-04
Epoch 144/500
1/1  0s 20ms/step - loss: 5.9564e-04
Epoch 145/500
1/1  0s 16ms/step - loss: 5.8340e-04
Epoch 146/500
1/1  0s 14ms/step - loss: 5.7141e-04
Epoch 147/500
1/1  0s 14ms/step - loss: 5.5968e-04
Epoch 148/500
1/1  0s 13ms/step - loss: 5.4818e-04
Epoch 149/500
1/1  0s 13ms/step - loss: 5.3692e-04
Epoch 150/500
1/1  0s 14ms/step - loss: 5.2589e-04

Epoch 151/500
1/1  0s 14ms/step - loss: 5.1509e-04
Epoch 152/500
1/1  0s 14ms/step - loss: 5.0451e-04
Epoch 153/500
1/1  0s 13ms/step - loss: 4.9415e-04
Epoch 154/500
1/1  0s 15ms/step - loss: 4.8400e-04
Epoch 155/500
1/1  0s 14ms/step - loss: 4.7406e-04
Epoch 156/500
1/1  0s 14ms/step - loss: 4.6432e-04
Epoch 157/500
1/1  0s 15ms/step - loss: 4.5478e-04
Epoch 158/500
1/1  0s 14ms/step - loss: 4.4544e-04
Epoch 159/500
1/1  0s 15ms/step - loss: 4.3629e-04
Epoch 160/500
1/1  0s 14ms/step - loss: 4.2733e-04
Epoch 161/500
1/1  0s 14ms/step - loss: 4.1855e-04
Epoch 162/500
1/1  0s 15ms/step - loss: 4.0996e-04
Epoch 163/500
1/1  0s 14ms/step - loss: 4.0153e-04
Epoch 164/500
1/1  0s 14ms/step - loss: 3.9329e-04
Epoch 165/500
1/1  0s 14ms/step - loss: 3.8521e-04
Epoch 166/500
1/1  0s 14ms/step - loss: 3.7730e-04
Epoch 167/500
1/1  0s 16ms/step - loss: 3.6955e-04
Epoch 168/500
1/1  0s 14ms/step - loss: 3.6195e-04
Epoch 169/500
1/1  0s 15ms/step - loss: 3.5452e-04
Epoch 170/500
1/1  0s 13ms/step - loss: 3.4724e-04
Epoch 171/500
1/1  0s 13ms/step - loss: 3.4010e-04
Epoch 172/500
1/1  0s 14ms/step - loss: 3.3312e-04
Epoch 173/500
1/1  0s 14ms/step - loss: 3.2627e-04
Epoch 174/500
1/1  0s 14ms/step - loss: 3.1957e-04
Epoch 175/500
1/1  0s 14ms/step - loss: 3.1301e-04
Epoch 176/500
1/1  0s 14ms/step - loss: 3.0658e-04
Epoch 177/500
1/1  0s 14ms/step - loss: 3.0028e-04
Epoch 178/500
1/1  0s 13ms/step - loss: 2.9412e-04
Epoch 179/500
1/1  0s 14ms/step - loss: 2.8808e-04
Epoch 180/500
1/1  0s 15ms/step - loss: 2.8216e-04

Epoch 181/500
1/1  0s 13ms/step - loss: 2.7636e-04
Epoch 182/500
1/1  0s 17ms/step - loss: 2.7068e-04
Epoch 183/500
1/1  0s 19ms/step - loss: 2.6513e-04
Epoch 184/500
1/1  0s 15ms/step - loss: 2.5968e-04
Epoch 185/500
1/1  0s 14ms/step - loss: 2.5435e-04
Epoch 186/500
1/1  0s 13ms/step - loss: 2.4912e-04
Epoch 187/500
1/1  0s 14ms/step - loss: 2.4400e-04
Epoch 188/500
1/1  0s 13ms/step - loss: 2.3899e-04
Epoch 189/500
1/1  0s 13ms/step - loss: 2.3408e-04
Epoch 190/500
1/1  0s 14ms/step - loss: 2.2928e-04
Epoch 191/500
1/1  0s 15ms/step - loss: 2.2456e-04
Epoch 192/500
1/1  0s 14ms/step - loss: 2.1996e-04
Epoch 193/500
1/1  0s 14ms/step - loss: 2.1543e-04
Epoch 194/500
1/1  0s 19ms/step - loss: 2.1101e-04
Epoch 195/500
1/1  0s 14ms/step - loss: 2.0668e-04
Epoch 196/500
1/1  0s 13ms/step - loss: 2.0243e-04
Epoch 197/500
1/1  0s 13ms/step - loss: 1.9827e-04
Epoch 198/500
1/1  0s 13ms/step - loss: 1.9420e-04
Epoch 199/500
1/1  0s 15ms/step - loss: 1.9021e-04
Epoch 200/500
1/1  0s 14ms/step - loss: 1.8631e-04
Epoch 201/500
1/1  0s 14ms/step - loss: 1.8248e-04
Epoch 202/500
1/1  0s 13ms/step - loss: 1.7873e-04
Epoch 203/500
1/1  0s 14ms/step - loss: 1.7506e-04
Epoch 204/500
1/1  0s 13ms/step - loss: 1.7146e-04
Epoch 205/500
1/1  0s 14ms/step - loss: 1.6794e-04
Epoch 206/500
1/1  0s 14ms/step - loss: 1.6449e-04
Epoch 207/500
1/1  0s 14ms/step - loss: 1.6111e-04
Epoch 208/500
1/1  0s 13ms/step - loss: 1.5780e-04
Epoch 209/500
1/1  0s 14ms/step - loss: 1.5456e-04
Epoch 210/500
1/1  0s 15ms/step - loss: 1.5139e-04

Epoch 211/500
1/1  0s 15ms/step - loss: 1.4828e-04
Epoch 212/500
1/1  0s 16ms/step - loss: 1.4523e-04
Epoch 213/500
1/1  0s 15ms/step - loss: 1.4225e-04
Epoch 214/500
1/1  0s 13ms/step - loss: 1.3933e-04
Epoch 215/500
1/1  0s 14ms/step - loss: 1.3647e-04
Epoch 216/500
1/1  0s 16ms/step - loss: 1.3366e-04
Epoch 217/500
1/1  0s 15ms/step - loss: 1.3092e-04
Epoch 218/500
1/1  0s 14ms/step - loss: 1.2823e-04
Epoch 219/500
1/1  0s 15ms/step - loss: 1.2559e-04
Epoch 220/500
1/1  0s 13ms/step - loss: 1.2301e-04
Epoch 221/500
1/1  0s 14ms/step - loss: 1.2049e-04
Epoch 222/500
1/1  0s 16ms/step - loss: 1.1802e-04
Epoch 223/500
1/1  0s 14ms/step - loss: 1.1559e-04
Epoch 224/500
1/1  0s 14ms/step - loss: 1.1322e-04
Epoch 225/500
1/1  0s 14ms/step - loss: 1.1089e-04
Epoch 226/500
1/1  0s 14ms/step - loss: 1.0861e-04
Epoch 227/500
1/1  0s 14ms/step - loss: 1.0638e-04
Epoch 228/500
1/1  0s 14ms/step - loss: 1.0420e-04
Epoch 229/500
1/1  0s 14ms/step - loss: 1.0206e-04
Epoch 230/500
1/1  0s 14ms/step - loss: 9.9959e-05
Epoch 231/500
1/1  0s 14ms/step - loss: 9.7906e-05
Epoch 232/500
1/1  0s 13ms/step - loss: 9.5896e-05
Epoch 233/500
1/1  0s 18ms/step - loss: 9.3925e-05
Epoch 234/500
1/1  0s 14ms/step - loss: 9.1996e-05
Epoch 235/500
1/1  0s 15ms/step - loss: 9.0106e-05
Epoch 236/500
1/1  0s 13ms/step - loss: 8.8255e-05
Epoch 237/500
1/1  0s 14ms/step - loss: 8.6443e-05
Epoch 238/500
1/1  0s 13ms/step - loss: 8.4667e-05
Epoch 239/500
1/1  0s 18ms/step - loss: 8.2928e-05
Epoch 240/500
1/1  0s 14ms/step - loss: 8.1224e-05

Epoch 241/500
1/1  0s 13ms/step - loss: 7.9556e-05
Epoch 242/500
1/1  0s 15ms/step - loss: 7.7922e-05
Epoch 243/500
1/1  0s 15ms/step - loss: 7.6322e-05
Epoch 244/500
1/1  0s 14ms/step - loss: 7.4753e-05
Epoch 245/500
1/1  0s 14ms/step - loss: 7.3218e-05
Epoch 246/500
1/1  0s 15ms/step - loss: 7.1716e-05
Epoch 247/500
1/1  0s 13ms/step - loss: 7.0243e-05
Epoch 248/500
1/1  0s 14ms/step - loss: 6.8799e-05
Epoch 249/500
1/1  0s 14ms/step - loss: 6.7386e-05
Epoch 250/500
1/1  0s 15ms/step - loss: 6.6002e-05
Epoch 251/500
1/1  0s 15ms/step - loss: 6.4647e-05
Epoch 252/500
1/1  0s 14ms/step - loss: 6.3318e-05
Epoch 253/500
1/1  0s 14ms/step - loss: 6.2018e-05
Epoch 254/500
1/1  0s 18ms/step - loss: 6.0743e-05
Epoch 255/500
1/1  0s 15ms/step - loss: 5.9496e-05
Epoch 256/500
1/1  0s 15ms/step - loss: 5.8275e-05
Epoch 257/500
1/1  0s 14ms/step - loss: 5.7077e-05
Epoch 258/500
1/1  0s 14ms/step - loss: 5.5905e-05
Epoch 259/500
1/1  0s 13ms/step - loss: 5.4756e-05
Epoch 260/500
1/1  0s 13ms/step - loss: 5.3631e-05
Epoch 261/500
1/1  0s 14ms/step - loss: 5.2529e-05
Epoch 262/500
1/1  0s 14ms/step - loss: 5.1450e-05
Epoch 263/500
1/1  0s 13ms/step - loss: 5.0394e-05
Epoch 264/500
1/1  0s 13ms/step - loss: 4.9357e-05
Epoch 265/500
1/1  0s 13ms/step - loss: 4.8345e-05
Epoch 266/500
1/1  0s 14ms/step - loss: 4.7351e-05
Epoch 267/500
1/1  0s 14ms/step - loss: 4.6378e-05
Epoch 268/500
1/1  0s 14ms/step - loss: 4.5425e-05
Epoch 269/500
1/1  0s 13ms/step - loss: 4.4493e-05
Epoch 270/500
1/1  0s 13ms/step - loss: 4.3578e-05

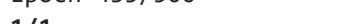
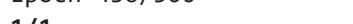
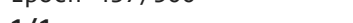
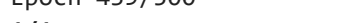
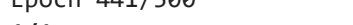
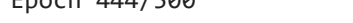
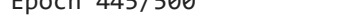
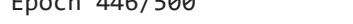
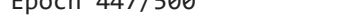
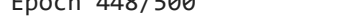
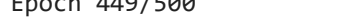
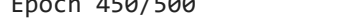
Epoch 271/500
1/1  0s 14ms/step - loss: 4.2685e-05
Epoch 272/500
1/1  0s 13ms/step - loss: 4.1807e-05
Epoch 273/500
1/1  0s 14ms/step - loss: 4.0948e-05
Epoch 274/500
1/1  0s 13ms/step - loss: 4.0107e-05
Epoch 275/500
1/1  0s 14ms/step - loss: 3.9284e-05
Epoch 276/500
1/1  0s 14ms/step - loss: 3.8476e-05
Epoch 277/500
1/1  0s 13ms/step - loss: 3.7686e-05
Epoch 278/500
1/1  0s 14ms/step - loss: 3.6911e-05
Epoch 279/500
1/1  0s 13ms/step - loss: 3.6153e-05
Epoch 280/500
1/1  0s 13ms/step - loss: 3.5410e-05
Epoch 281/500
1/1  0s 13ms/step - loss: 3.4684e-05
Epoch 282/500
1/1  0s 12ms/step - loss: 3.3971e-05
Epoch 283/500
1/1  0s 13ms/step - loss: 3.3273e-05
Epoch 284/500
1/1  0s 14ms/step - loss: 3.2589e-05
Epoch 285/500
1/1  0s 13ms/step - loss: 3.1920e-05
Epoch 286/500
1/1  0s 13ms/step - loss: 3.1265e-05
Epoch 287/500
1/1  0s 14ms/step - loss: 3.0622e-05
Epoch 288/500
1/1  0s 14ms/step - loss: 2.9994e-05
Epoch 289/500
1/1  0s 13ms/step - loss: 2.9378e-05
Epoch 290/500
1/1  0s 14ms/step - loss: 2.8775e-05
Epoch 291/500
1/1  0s 14ms/step - loss: 2.8184e-05
Epoch 292/500
1/1  0s 14ms/step - loss: 2.7605e-05
Epoch 293/500
1/1  0s 13ms/step - loss: 2.7038e-05
Epoch 294/500
1/1  0s 14ms/step - loss: 2.6483e-05
Epoch 295/500
1/1  0s 14ms/step - loss: 2.5940e-05
Epoch 296/500
1/1  0s 13ms/step - loss: 2.5407e-05
Epoch 297/500
1/1  0s 18ms/step - loss: 2.4886e-05
Epoch 298/500
1/1  0s 14ms/step - loss: 2.4374e-05
Epoch 299/500
1/1  0s 13ms/step - loss: 2.3873e-05
Epoch 300/500
1/1  0s 13ms/step - loss: 2.3383e-05

Epoch 301/500
1/1  0s 14ms/step - loss: 2.2903e-05
Epoch 302/500
1/1  0s 14ms/step - loss: 2.2432e-05
Epoch 303/500
1/1  0s 13ms/step - loss: 2.1971e-05
Epoch 304/500
1/1  0s 14ms/step - loss: 2.1520e-05
Epoch 305/500
1/1  0s 13ms/step - loss: 2.1078e-05
Epoch 306/500
1/1  0s 15ms/step - loss: 2.0645e-05
Epoch 307/500
1/1  0s 13ms/step - loss: 2.0220e-05
Epoch 308/500
1/1  0s 13ms/step - loss: 1.9806e-05
Epoch 309/500
1/1  0s 16ms/step - loss: 1.9399e-05
Epoch 310/500
1/1  0s 17ms/step - loss: 1.9000e-05
Epoch 311/500
1/1  0s 13ms/step - loss: 1.8610e-05
Epoch 312/500
1/1  0s 14ms/step - loss: 1.8228e-05
Epoch 313/500
1/1  0s 13ms/step - loss: 1.7853e-05
Epoch 314/500
1/1  0s 14ms/step - loss: 1.7487e-05
Epoch 315/500
1/1  0s 13ms/step - loss: 1.7127e-05
Epoch 316/500
1/1  0s 13ms/step - loss: 1.6775e-05
Epoch 317/500
1/1  0s 13ms/step - loss: 1.6431e-05
Epoch 318/500
1/1  0s 13ms/step - loss: 1.6093e-05
Epoch 319/500
1/1  0s 13ms/step - loss: 1.5763e-05
Epoch 320/500
1/1  0s 14ms/step - loss: 1.5439e-05
Epoch 321/500
1/1  0s 12ms/step - loss: 1.5122e-05
Epoch 322/500
1/1  0s 13ms/step - loss: 1.4811e-05
Epoch 323/500
1/1  0s 14ms/step - loss: 1.4507e-05
Epoch 324/500
1/1  0s 15ms/step - loss: 1.4210e-05
Epoch 325/500
1/1  0s 14ms/step - loss: 1.3918e-05
Epoch 326/500
1/1  0s 13ms/step - loss: 1.3632e-05
Epoch 327/500
1/1  0s 14ms/step - loss: 1.3352e-05
Epoch 328/500
1/1  0s 13ms/step - loss: 1.3077e-05
Epoch 329/500
1/1  0s 13ms/step - loss: 1.2809e-05
Epoch 330/500
1/1  0s 12ms/step - loss: 1.2546e-05

Epoch 331/500
1/1  0s 13ms/step - loss: 1.2288e-05
Epoch 332/500
1/1  0s 14ms/step - loss: 1.2035e-05
Epoch 333/500
1/1  0s 13ms/step - loss: 1.1788e-05
Epoch 334/500
1/1  0s 14ms/step - loss: 1.1546e-05
Epoch 335/500
1/1  0s 14ms/step - loss: 1.1308e-05
Epoch 336/500
1/1  0s 14ms/step - loss: 1.1076e-05
Epoch 337/500
1/1  0s 13ms/step - loss: 1.0848e-05
Epoch 338/500
1/1  0s 13ms/step - loss: 1.0625e-05
Epoch 339/500
1/1  0s 14ms/step - loss: 1.0407e-05
Epoch 340/500
1/1  0s 13ms/step - loss: 1.0193e-05
Epoch 341/500
1/1  0s 13ms/step - loss: 9.9841e-06
Epoch 342/500
1/1  0s 13ms/step - loss: 9.7791e-06
Epoch 343/500
1/1  0s 13ms/step - loss: 9.5779e-06
Epoch 344/500
1/1  0s 13ms/step - loss: 9.3817e-06
Epoch 345/500
1/1  0s 13ms/step - loss: 9.1887e-06
Epoch 346/500
1/1  0s 14ms/step - loss: 8.9998e-06
Epoch 347/500
1/1  0s 13ms/step - loss: 8.8151e-06
Epoch 348/500
1/1  0s 14ms/step - loss: 8.6341e-06
Epoch 349/500
1/1  0s 13ms/step - loss: 8.4575e-06
Epoch 350/500
1/1  0s 13ms/step - loss: 8.2833e-06
Epoch 351/500
1/1  0s 13ms/step - loss: 8.1137e-06
Epoch 352/500
1/1  0s 13ms/step - loss: 7.9463e-06
Epoch 353/500
1/1  0s 14ms/step - loss: 7.7836e-06
Epoch 354/500
1/1  0s 13ms/step - loss: 7.6237e-06
Epoch 355/500
1/1  0s 13ms/step - loss: 7.4672e-06
Epoch 356/500
1/1  0s 13ms/step - loss: 7.3139e-06
Epoch 357/500
1/1  0s 14ms/step - loss: 7.1628e-06
Epoch 358/500
1/1  0s 14ms/step - loss: 7.0164e-06
Epoch 359/500
1/1  0s 13ms/step - loss: 6.8720e-06
Epoch 360/500
1/1  0s 15ms/step - loss: 6.7308e-06

Epoch 361/500
1/1  0s 13ms/step - loss: 6.5925e-06
Epoch 362/500
1/1  0s 13ms/step - loss: 6.4574e-06
Epoch 363/500
1/1  0s 13ms/step - loss: 6.3249e-06
Epoch 364/500
1/1  0s 13ms/step - loss: 6.1948e-06
Epoch 365/500
1/1  0s 13ms/step - loss: 6.0674e-06
Epoch 366/500
1/1  0s 13ms/step - loss: 5.9432e-06
Epoch 367/500
1/1  0s 14ms/step - loss: 5.8210e-06
Epoch 368/500
1/1  0s 16ms/step - loss: 5.7015e-06
Epoch 369/500
1/1  0s 18ms/step - loss: 5.5847e-06
Epoch 370/500
1/1  0s 13ms/step - loss: 5.4694e-06
Epoch 371/500
1/1  0s 13ms/step - loss: 5.3573e-06
Epoch 372/500
1/1  0s 14ms/step - loss: 5.2471e-06
Epoch 373/500
1/1  0s 13ms/step - loss: 5.1396e-06
Epoch 374/500
1/1  0s 14ms/step - loss: 5.0342e-06
Epoch 375/500
1/1  0s 17ms/step - loss: 4.9305e-06
Epoch 376/500
1/1  0s 13ms/step - loss: 4.8294e-06
Epoch 377/500
1/1  0s 13ms/step - loss: 4.7301e-06
Epoch 378/500
1/1  0s 14ms/step - loss: 4.6327e-06
Epoch 379/500
1/1  0s 13ms/step - loss: 4.5374e-06
Epoch 380/500
1/1  0s 12ms/step - loss: 4.4445e-06
Epoch 381/500
1/1  0s 14ms/step - loss: 4.3530e-06
Epoch 382/500
1/1  0s 12ms/step - loss: 4.2633e-06
Epoch 383/500
1/1  0s 14ms/step - loss: 4.1761e-06
Epoch 384/500
1/1  0s 13ms/step - loss: 4.0902e-06
Epoch 385/500
1/1  0s 14ms/step - loss: 4.0063e-06
Epoch 386/500
1/1  0s 13ms/step - loss: 3.9239e-06
Epoch 387/500
1/1  0s 13ms/step - loss: 3.8431e-06
Epoch 388/500
1/1  0s 14ms/step - loss: 3.7644e-06
Epoch 389/500
1/1  0s 13ms/step - loss: 3.6872e-06
Epoch 390/500
1/1  0s 14ms/step - loss: 3.6109e-06

Epoch 391/500
1/1  0s 14ms/step - loss: 3.5369e-06
Epoch 392/500
1/1  0s 13ms/step - loss: 3.4642e-06
Epoch 393/500
1/1  0s 13ms/step - loss: 3.3932e-06
Epoch 394/500
1/1  0s 14ms/step - loss: 3.3236e-06
Epoch 395/500
1/1  0s 13ms/step - loss: 3.2554e-06
Epoch 396/500
1/1  0s 14ms/step - loss: 3.1884e-06
Epoch 397/500
1/1  0s 14ms/step - loss: 3.1230e-06
Epoch 398/500
1/1  0s 13ms/step - loss: 3.0586e-06
Epoch 399/500
1/1  0s 13ms/step - loss: 2.9960e-06
Epoch 400/500
1/1  0s 14ms/step - loss: 2.9347e-06
Epoch 401/500
1/1  0s 13ms/step - loss: 2.8740e-06
Epoch 402/500
1/1  0s 12ms/step - loss: 2.8150e-06
Epoch 403/500
1/1  0s 14ms/step - loss: 2.7574e-06
Epoch 404/500
1/1  0s 13ms/step - loss: 2.7006e-06
Epoch 405/500
1/1  0s 13ms/step - loss: 2.6451e-06
Epoch 406/500
1/1  0s 14ms/step - loss: 2.5908e-06
Epoch 407/500
1/1  0s 13ms/step - loss: 2.5376e-06
Epoch 408/500
1/1  0s 14ms/step - loss: 2.4855e-06
Epoch 409/500
1/1  0s 14ms/step - loss: 2.4345e-06
Epoch 410/500
1/1  0s 13ms/step - loss: 2.3844e-06
Epoch 411/500
1/1  0s 13ms/step - loss: 2.3354e-06
Epoch 412/500
1/1  0s 13ms/step - loss: 2.2874e-06
Epoch 413/500
1/1  0s 14ms/step - loss: 2.2404e-06
Epoch 414/500
1/1  0s 13ms/step - loss: 2.1945e-06
Epoch 415/500
1/1  0s 15ms/step - loss: 2.1497e-06
Epoch 416/500
1/1  0s 13ms/step - loss: 2.1053e-06
Epoch 417/500
1/1  0s 13ms/step - loss: 2.0619e-06
Epoch 418/500
1/1  0s 13ms/step - loss: 2.0197e-06
Epoch 419/500
1/1  0s 12ms/step - loss: 1.9783e-06
Epoch 420/500
1/1  0s 14ms/step - loss: 1.9377e-06

Epoch 421/500
1/1  0s 13ms/step - loss: 1.8979e-06
Epoch 422/500
1/1  0s 12ms/step - loss: 1.8587e-06
Epoch 423/500
1/1  0s 14ms/step - loss: 1.8206e-06
Epoch 424/500
1/1  0s 14ms/step - loss: 1.7831e-06
Epoch 425/500
1/1  0s 14ms/step - loss: 1.7465e-06
Epoch 426/500
1/1  0s 14ms/step - loss: 1.7107e-06
Epoch 427/500
1/1  0s 16ms/step - loss: 1.6756e-06
Epoch 428/500
1/1  0s 13ms/step - loss: 1.6413e-06
Epoch 429/500
1/1  0s 13ms/step - loss: 1.6074e-06
Epoch 430/500
1/1  0s 12ms/step - loss: 1.5743e-06
Epoch 431/500
1/1  0s 14ms/step - loss: 1.5423e-06
Epoch 432/500
1/1  0s 14ms/step - loss: 1.5104e-06
Epoch 433/500
1/1  0s 12ms/step - loss: 1.4795e-06
Epoch 434/500
1/1  0s 15ms/step - loss: 1.4489e-06
Epoch 435/500
1/1  0s 14ms/step - loss: 1.4192e-06
Epoch 436/500
1/1  0s 14ms/step - loss: 1.3901e-06
Epoch 437/500
1/1  0s 13ms/step - loss: 1.3614e-06
Epoch 438/500
1/1  0s 12ms/step - loss: 1.3334e-06
Epoch 439/500
1/1  0s 14ms/step - loss: 1.3062e-06
Epoch 440/500
1/1  0s 13ms/step - loss: 1.2792e-06
Epoch 441/500
1/1  0s 14ms/step - loss: 1.2532e-06
Epoch 442/500
1/1  0s 14ms/step - loss: 1.2272e-06
Epoch 443/500
1/1  0s 13ms/step - loss: 1.2022e-06
Epoch 444/500
1/1  0s 14ms/step - loss: 1.1775e-06
Epoch 445/500
1/1  0s 14ms/step - loss: 1.1533e-06
Epoch 446/500
1/1  0s 13ms/step - loss: 1.1297e-06
Epoch 447/500
1/1  0s 13ms/step - loss: 1.1063e-06
Epoch 448/500
1/1  0s 12ms/step - loss: 1.0837e-06
Epoch 449/500
1/1  0s 13ms/step - loss: 1.0614e-06
Epoch 450/500
1/1  0s 13ms/step - loss: 1.0397e-06

Epoch 451/500
1/1  0s 14ms/step - loss: 1.0183e-06
Epoch 452/500
1/1  0s 12ms/step - loss: 9.9745e-07
Epoch 453/500
1/1  0s 13ms/step - loss: 9.7677e-07
Epoch 454/500
1/1  0s 13ms/step - loss: 9.5680e-07
Epoch 455/500
1/1  0s 13ms/step - loss: 9.3719e-07
Epoch 456/500
1/1  0s 14ms/step - loss: 9.1790e-07
Epoch 457/500
1/1  0s 13ms/step - loss: 8.9900e-07
Epoch 458/500
1/1  0s 13ms/step - loss: 8.8060e-07
Epoch 459/500
1/1  0s 13ms/step - loss: 8.6248e-07
Epoch 460/500
1/1  0s 13ms/step - loss: 8.4478e-07
Epoch 461/500
1/1  0s 13ms/step - loss: 8.2738e-07
Epoch 462/500
1/1  0s 14ms/step - loss: 8.1048e-07
Epoch 463/500
1/1  0s 13ms/step - loss: 7.9362e-07
Epoch 464/500
1/1  0s 14ms/step - loss: 7.7746e-07
Epoch 465/500
1/1  0s 13ms/step - loss: 7.6143e-07
Epoch 466/500
1/1  0s 13ms/step - loss: 7.4577e-07
Epoch 467/500
1/1  0s 13ms/step - loss: 7.3059e-07
Epoch 468/500
1/1  0s 13ms/step - loss: 7.1559e-07
Epoch 469/500
1/1  0s 13ms/step - loss: 7.0095e-07
Epoch 470/500
1/1  0s 12ms/step - loss: 6.8643e-07
Epoch 471/500
1/1  0s 13ms/step - loss: 6.7245e-07
Epoch 472/500
1/1  0s 14ms/step - loss: 6.5871e-07
Epoch 473/500
1/1  0s 13ms/step - loss: 6.4501e-07
Epoch 474/500
1/1  0s 13ms/step - loss: 6.3198e-07
Epoch 475/500
1/1  0s 13ms/step - loss: 6.1892e-07
Epoch 476/500
1/1  0s 13ms/step - loss: 6.0626e-07
Epoch 477/500
1/1  0s 14ms/step - loss: 5.9377e-07
Epoch 478/500
1/1  0s 13ms/step - loss: 5.8162e-07
Epoch 479/500
1/1  0s 13ms/step - loss: 5.6964e-07
Epoch 480/500
1/1  0s 13ms/step - loss: 5.5799e-07

```
Epoch 481/500
1/1  0s 12ms/step - loss: 5.4645e-07
Epoch 482/500
1/1  0s 14ms/step - loss: 5.3525e-07
Epoch 483/500
1/1  0s 17ms/step - loss: 5.2417e-07
Epoch 484/500
1/1  0s 15ms/step - loss: 5.1345e-07
Epoch 485/500
1/1  0s 13ms/step - loss: 5.0290e-07
Epoch 486/500
1/1  0s 13ms/step - loss: 4.9268e-07
Epoch 487/500
1/1  0s 13ms/step - loss: 4.8245e-07
Epoch 488/500
1/1  0s 14ms/step - loss: 4.7266e-07
Epoch 489/500
1/1  0s 14ms/step - loss: 4.6284e-07
Epoch 490/500
1/1  0s 13ms/step - loss: 4.5331e-07
Epoch 491/500
1/1  0s 15ms/step - loss: 4.4393e-07
Epoch 492/500
1/1  0s 13ms/step - loss: 4.3490e-07
Epoch 493/500
1/1  0s 13ms/step - loss: 4.2592e-07
Epoch 494/500
1/1  0s 14ms/step - loss: 4.1727e-07
Epoch 495/500
1/1  0s 13ms/step - loss: 4.0871e-07
Epoch 496/500
1/1  0s 14ms/step - loss: 4.0026e-07
Epoch 497/500
1/1  0s 13ms/step - loss: 3.9201e-07
Epoch 498/500
1/1  0s 15ms/step - loss: 3.8392e-07
Epoch 499/500
1/1  0s 13ms/step - loss: 3.7605e-07
Epoch 500/500
1/1  0s 13ms/step - loss: 3.6836e-07
1/1  0s 22ms/step
[[31.001772]]
```

In []: